LASARE SAMARTZIDIS

lasare@web.de ◇ LinkedIn ◇ personal website

PROFESSIONAL SUMMARY

Ecological and complexity economist (PhD, magna cum laude, TU Dresden 2026) working on production networks, regional development and inequality, functional income distribution and subjective well-being. Methods span network and input-output analysis, agent-based modelling, explainable machine learning and LLM-based analysis of scientific literature, delivered in reproducible, open-science workflows with data and code archived on Zenodo. Long-standing experience coordinating inter- and transdisciplinary research across institutions, and in communicating results through dashboards, teaching and 18 conference presentations.

KEY SKILLS

Networks & Input-Output Analysis: Production networks and their spatial structure; complex network methods (NetSci 2023); regional and multi-regional input-output models (WIOD, IRIOs); induced demand effects

Agent-Based & Complexity Modelling: Economy as a complex adaptive system; ABM training (Sant'Anna School, Pisa); LSD

Machine Learning & LLMs: Random forests, XGBoost, LASSO, cross-validation; explainable ML (ALE, Friedman's H); LLM pipelines: PDF extraction, chunking, embeddings, RAG, structured JSON extraction

Econometrics & Spatial Analysis: Regression and panel data; geospatial analysis and mapping (GIS, grid data, travel-time accessibility)

Reproducible Research: Git/GitHub, CI with GitHub Actions, tested data pipelines (dbt, DuckDB), Quarto, open data and code on Zenodo

Research Coordination & Communication: Two scientific coordinator roles in Leibniz networks; workshops, general assemblies, seminar series; Shiny dashboards; Leibniz "Book a Scientist"

EDUCATION

Technische Universität Dresden — *Defended July 2026*
PhD in Economics (Dr. rer. pol.), magna cum laude — *Dresden, Germany*

- Dissertation: "Sustainable Regional Development: A Complexity Economics Analysis" supervisor: Prof. Dr. Artem Korzhenevych
- Coursework: Macroeconomics of Climate Change, Econometrics, Causal Inference in Applied Economics and Business Economics, Agent-Based Modelling, Research Data Management, Academic Writing, Academic Presentations

Kiel University — *2019*
Master of Science in Economics — *Kiel, Germany*

- Thesis: "Who finds what repugnant and why?"; supervisors: Prof. Dr. Dennis Snower & Prof. Dr. Ralph Hertwig
- Resulted in two publications with the Max Planck Institute for Human Development

Friedrich Schiller University Jena — *2014*
Bachelor of Science in Business and Economics — *Jena, Germany*

EXPERIENCE

Leibniz Institute of Ecological Urban and Regional Development (IÖR) — *11/2020 – 03/2026*, *Dresden*

Scientific Coordinator, Leibniz Lab "Systemic Sustainability" (05/2024 – 03/2026)

- Initiated, led and coordinated a cross-institutional team producing inter- and transdisciplinary research (15 people from 11 institutions, working remotely), in close alignment with the Lab's spokesperson and steering group
- Contributed to the conceptual and methodological development of the Lab; ensured knowledge flows with national and international research partners and societal stakeholders
- Organised hybrid General Assemblies and workshops; initiated an ongoing seminar series

Scientific Coordinator, Leibniz Research Network "Earth & Societies" (07/2023 – 04/2024)

- Organised and documented network meetings; managed and secured information flows between partner institutions

Doctoral Researcher (11/2020 – 03/2026)

- Independently conceptualised, conducted and published research on sustainable regional development: production networks, regional synchronisation, functional income distribution, subjective well-being
- Developed reproducible, code-first workflows with Git for highly data-driven research; data and code archived on Zenodo

WIG2 – Scientific Institute for Health Economics and Health System Research *02/2022 – 07/2023*
Researcher *Leipzig*

- Accessibility of hospices in Saxony: prepared georeferenced data on a 1 km grid, computed population shares reaching hospices by public transport within given travel times, and mapped the results (GIS)
- Initiated and built a Shiny dashboard for interactive exploration of the results; built an internal R package for the institute

Fraunhofer IMW – Center for Economics of Materials (CEM) *07/2019 – 04/2021*
Researcher *Halle (Saale)*

- Contributed to a report on the potential of a green regional gas industry in Central Germany
- Co-developed a regional input-output model; coded a Shiny app estimating regional induced demand effects of industrial investments
- Led student assistants

PUBLICATIONS

Samartzidis, L. (2025). Synchronisation of regional development – the role of clusters in the regional production network. *Spatial Economic Analysis*. [DOI] [Data & Code]

Samartzidis, L., Quatrosi, M., & von Dulong, A. (2025). Gender and age matter! Identifying important predictors for subjective well-being using machine learning methods. *Social Indicators Research*, 179(2), 955–978. [DOI] [Data & Code]

De Castro Mazarro, A., Joshi, N., & **Samartzidis, L.** (2024). Disrupting comfort: From low-carbon to low-impact cities. *Raumforschung und Raumordnung | Spatial Research and Planning*, 82(4), 289–293. [DOI]

Leuker, C., **Samartzidis, L.**, & Hertwig, R. (2021). What makes a market transaction morally repugnant? *Cognition*, 212, 104644. [DOI]

Leuker, C., **Samartzidis, L.**, Hertwig, R., & Pleskac, T. J. (2020). When money talks: Judging risk and coercion in high-paying clinical trials. *PLoS ONE*, 15(1), e0227898. [DOI]

MANUSCRIPTS

Samartzidis, L., Mundt, P., & Schulz-Gephard, J. Input specificity and labour's bargaining power: A production tree approach to functional income distribution. *Economic Modelling* (under revision). [Working Paper] [Data & Code]

Samartzidis, L., & Van Eynde, R. Are macroeconomic agent-based models ready for post-growth policies? A systematic review with large language models. Manuscript in preparation. [Preprint]

OTHER PUBLICATIONS

Samartzidis, L. (2026). *Sustainable Regional Development: A Complexity Economics Analysis.* Doctoral thesis, Technische Universität Dresden. [URN]

Rösel, F., & **Samartzidis, L.** (2018). Wert- statt Geldmangel: AfD triumphiert im Osten dort, wo es an Toleranz und Vertrauen in die Politik fehlt. *ifo Dresden berichtet*, 25(3), 9–13.

RESEARCH SOFTWARE & PROJECTS

LLM pipeline for a systematic literature review (with R. Van Eynde) *2025 – 2026*
R, Python, Scopus API, PyMuPDF, embeddings, RAG, open-weight LLMs *Website post*

- Four-stage, human-in-the-loop pipeline turning a Scopus search into a structured database, replacing the manual coding of about 2,500 data points (approx. 40 features in 62 full texts)
- LLM relevance scoring of abstracts with manual review of borderline cases; PDF extraction, chunking and embeddings; structured extraction of 41 model elements per paper as validated JSON with supporting excerpts; open-weight LLMs (Llama 3.1, Mistral Large, GPT-OSS-120B) run reproducibly at temperature 0

OECD Regional Well-Being: tested data pipeline *2026*
dbt, DuckDB, SQL, Python, scikit-learn, pytest, GitHub Actions *Website post*

- Rebuilt the data preparation and ML analysis of the *Social Indicators Research* article as a version-controlled, CI-tested pipeline that reproduces the published dataset exactly (388 regions, 135 indicators)
- Declarative data tests surfaced four previously unnoticed data issues; Python implementations of ALE and Friedman's H-statistic verified against analytical cases

TEACHING

2026: *Lecturer, Introductory Course in Economics*, University of Applied Sciences (HTW) Dresden. Built an interactive teaching tool with generative AI (Vanilla JS, Canvas) for exploring economic models (efficiency wages, markets, tax incidence)

2025: *Guest lecture*, University of Duisburg-Essen, online

PRESENTATIONS

2025: *11th International Degrowth Conference / 18th Conference of the European Society for Ecological Economics*, Oslo (NO)

2024: *Winter Seminar, Gesellschaft für Regionalforschung (GfR)*, Igls (AT) [Programme]; *10th International Degrowth Conference / 15th Conference of the European Society for Ecological Economics*, Pontevedra (ES); *5th Pluralumni Workshop*, Duisburg; *28th Conference of the Forum for Macroeconomics and Macroeconomic Policies (FMM)*, Berlin [Programme]

2023: *Winter Seminar, Gesellschaft für Regionalforschung (GfR)*, Spital am Pyhrn (AT) [Programme]; *10th CGDE Doctoral Workshop*, Jena [Programme]; *NetSci Annual Conference*, Vienna (AT) [Programme]; *EAEPE Annual Conference*, Leeds (UK)

2022: *3rd Pluralumni Workshop*, Bielefeld; *LNRI Workshop on Regional Economic Inequalities*, Paris (FR) [Programme]

2021: *Advancing Indicators of Regional Structural Change, TechSpace Workshop*, online [Programme]; *27th Annual Conference, ISDRS*, online; *EAEPE Annual Conference*, online; *11th Summer Conference in Regional Science, Gesellschaft für Regionalforschung (GfR)*, online [Programme]; *2nd Pluralumni Workshop*, Bielefeld

2020: *Cognitive Economics Virtual Conference*, online [Summary]

2019: *Wittenberg Interdisciplinary Business Ethics Conference for Young Researchers (WIBEC)*, Wittenberg

FURTHER EDUCATION

Autumn School InfraTrain 2019: "Applied Econometrics in Environmental and Sustainability Economics", TU Berlin

Seasonal School 2022 on Agent-Based Models in Economics: Theory, Toolkit and Policy Laboratories, Sant'Anna School of Advanced Studies, Pisa

Central-German Doctoral Program Economics: Macroeconomics of Climate Change, Causal Inference in Applied Economics and Business Economics, Econometrics

Graduate Academy, TU Dresden: Academic Presentations, Research Data Management, Academic Writing, Third-Party Funding

LANGUAGES AND SOFTWARE

Languages	German (native), English (fluent), French (beginner)
Software	R (expert, 8+ years: tidyverse, tidymodels, Shiny, ggplot2), Python (pandas, scikit-learn, XGBoost, pytest), SQL (dbt, DuckDB), Git, GitHub Actions, Quarto, LaTeX, LSD (agent-based simulation), Claude Code

RESEARCH INTERESTS

Sustainability transformation research, ecological economics, complexity economics, production networks and input-output analysis, spatial and regional economics, structural change, agent-based modelling, subjective well-being, open science